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TFDA: A Time-Frequency Domain Adaptive Model for Multi-Source Multi-Task Anomaly Detection and Localization in Distribution Systems

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***Abstract*—Accurate anomaly detection and localization in distribution systems enable the timely event identification and the implementation of appropriate response strategies, which could minimize potential damage. However, many existing methods rely on extensive sensor data, requiring comprehensive measurement variables for data preprocessing and fusion, which limits their applicability in real-world scenarios. Moreover, prior methods often lack the necessary adaptive mechanisms to account for the variability of anomalies, which lead to inconsistent performance across tasks. To address these challenges, we propose a data-driven time-frequency domain adaptive model (TFDA) that incorporates both time-domain and frequency-domain features, only utilizing voltage magnitudes. The model introduces a spatial patch embedding block, which integrates sensor spatial information as graph priors and processes time-series data through patch-based operations to mitigate the impact of multi-resolution data. In the frequency-domain block, an adaptive thresholding technique is applied to eliminate less informative parts of the frequency spectrum. In the time-domain block, an adaptive weighting is utilized to balance the contributions of inter-channel and intra-channel anomaly information. We perform simulations on hardware-in-the-loop testbed to generate data that closely approximates real-world scenarios and demonstrate the superiority of our method over the baseline.**

***Index Terms*—Distribution system, anomaly detection, anomaly localization, data fusion, adaptive model**

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I. Introduction

THE stable operation of distribution systems is essential to ensuring the reliability and security of power supply. However, various anomalies such as short-circuit faults, cyber-attacks, and load jumps may occur during practical operations, which can significantly impact the normal functioning of the distribution system [1]. Different types of anomalies require specific response strategies to minimize potential losses. Hence, real-time monitoring and localization of anomalies are critical for operators to take prompt actions, which ensure the reliable operation of distribution system. With advancements in technology, distribution system have been increasingly equipped with a variety of advanced measurement devices capable of providing abundant real-time data [2]. These data present valuable opportunities for anomaly detection and localization, which enable effective data analysis to quickly identify and locate potential threats [3].

Initially, research on anomalies in distribution systems predominantly focused on addressing individual anomaly types. Among these, fault detection and localization emerged as key research areas. For distribution systems with clear and simple structures, physics-based methods, such as impedance-based approaches [4], as well as traveling wave and wavelet-based techniques [5], were proved to be highly effective in solving these challenges. However, as distribution systems became more complex with modernization, the need for advanced analysis of measurement data became increasingly apparent.

To tackle these complexities, some studies developed specialized solutions for specific fault scenarios. For instance, graph marking-based methods were proposed to localize feeder faults by integrating real-time data from fault indicators (FIs) with topological search algorithms [6]. Specialized time-domain techniques have been developed to localize faults by exploiting the transformer saturation effects, utilizing low-pass filters to extract the characteristic offset in phase currents for diagnostic purposes [7]. Trindade et al. [8] proposed a method that leveraged power flow calculations for data analysis. Key et al. [9] employed stacked denoising autoencoders to reliably discriminate transformer internal faults and Ngo et al. [10] developed a hybrid 1-D convolutional graph attention network to capture spatio-temporal features from system measurements for enhanced fault diagnostics. With the growing coupling

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of cyber and physical systems, researchers also increasingly focused on cyber attacks. Li et al. [11] proposed an improved spectral clustering-based network partitioning method that first estimated rough partitions and then determined precise locations. Lin et al. [12] introduced a novel edge-based federated learning framework, which demonstrated significant potential for detecting attacks in scenarios with unknown system parameters.

However, the wide range of anomalies in distribution systems makes it insufficient to focus solely on a single anomaly type to ensure system safety. Zhou et al. [13] considered multiple anomalies, including voltage sags, motor starts, and faults, and proposed a novel hidden-structure semi-supervised machine (HSM). They also developed a parametric dual optimization procedure to address the non-convex learning objective. Gholami et al. [14] integrated faults, false data injection attacks, and distributed energy resources (DER) switching by developing an ensemble method based on marginal bayesian maximum likelihood estimation (MB-MLE), which combined the results of various base detectors to identify anomalous events. Aligholian et al. [15] addressed faults and load jumps using an unsupervised graph representation learning method that leveraged the relative positions of sensors. Guo et al. [16] constructed features that can reflect the specific physical characteristics, and used convolutional neural networks (CNN) to distinguish between faults and cyber attacks. Vosughi et al. [17] proposed a comprehensive algorithmic workflow for analysing anomalied such as faults, DER switching, and load jumps. This workflow first detected anomalies using a prony-based dynamic window, classified them using koopman mode analysis (KMA), physical rules, and a weighted voting ensemble, and finally localized the anomalies using a weighted graph approach. Notably, most of the aforementioned studies relied on phasor measurement unit (PMU) data due to its comprehensive nature.

Given the cost constraints in distribution systems, the number of measurement devices was often limited in practical scenarios. Relying on a single type of information source did not provide a comprehensive view of the distribution system. Considering the wide variety of anomalies and the limited availability of sensors, integrating multi-source sensor data significantly enhanced situational awareness in these systems. Siamak et al. [18] proposed a sequential pattern mining approach for anomaly detection by utilizing data from PMUs, protection relays, and network event logs. Ahmed et al. [19] combined physical measurements from PMUs with network data from communication network logs. By merging these datasets through a weighted adjacency matrix, they introduced a deep graph learning framework to extract spatiotemporal features, which enabled the identification, localization, and classification of cyber-physical events. Gholami et al. [20] aggregated PMUs and relay data using an ensemble extended Kalman filter and performed classification using hierarchical agglomerative clustering. Similarly, Ganjkhani et al. [21] utilized data from PMUs and protection relays to estimate unknown variables in order to expand data features. Subsequently, long short-term memory (LSTM) networks were applied to achieve anomaly detection and localization.

However, it is important to note that most existing studies assumed the availability of a comprehensive set of sensor variables, including voltage magnitude, voltage phase angle, current magnitude, current phase angle, and frequency. These variables were typically processed through operations such as state estimation and feature expansion before performing subsequent detection and localization tasks. When the available variables were incomplete, many data processing operations became infeasible, rendering the methods unable to complete the intended tasks effectively. The comparative analysis between these specialized power system methods, focusing on data availability and anomaly types, is summarized in Table I.

Furthermore, while advanced time-series models, including FiLM [24], TimesNet [32], and ModernTCN [25], offer powerful feature extraction, they predominantly rely on a single informational domain (either temporal or spectral). In distribution systems, however, different anomalies exhibit non-uniform signatures; for instance, short-circuit faults cause sharp time-domain transients, while stealthy cyber-attacks are often more discernible through frequency-domain distortions. By neglecting the synergistic integration of both domains, existing methods fail to capture the multi-dimensional characteristics of diverse events. Moreover, without appropriate adaptive mechanisms to dynamically weight these features, these methods exhibit unstable performance across heterogeneous localization tasks, thereby failing to meet the safety and reliability standards required for distribution systems.

To address the aforementioned issues, this work proposes a Time-Frequency Domain Adaptive (TFDA) framework that relies exclusively on voltage magnitude. By extracting synergistic features from both the time and frequency domains, the model effectively captures the multi-dimensional signatures of diverse anomalies. To handle multi-source data with limited variables, we introduce a novel embedding approach. Furthermore, dual-domain adaptive operations are incorporated to ensure the framework meets the safety requirements for various anomaly localization tasks.

The main contributions of our work are as follows:

1) We develop a time-frequency domain adaptive model (TFDA) that enables highly reliable anomaly detection and localization utilizing only voltage magnitudes. This approach significantly reduces the dependency on multi-variable data, overcoming the practical deployment limitations faced by previous methods such as [20], [21].
2) To bridge the gap between heterogeneous sensor resolutions, we propose a spatial patch embedding module. By shifting from point-based data to patch-based analysis while integrating graph-based spatial priors, this module mitigates information imbalance and ensures the effective aggregation of multi-source measurement data.
3) We introduce domain-specific adaptive mechanisms to handle the distinct characteristics of diverse anomalies. The frequency-domain block employs an adaptive mask to prioritize informative spectral components, while the time-domain block utilizes learnable weighting to balance inter-variable topological relationships and intra-variable trends. These mechanisms enable stable and robust performance across diverse localization tasks, overcoming

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TABLE I: A comparison of our work with some of the latest research.

Related Work	Utilized Data				Considered Anomalies				
	I	II	III	IV	Fault	Relay Attack	Inverter Attack	DER Switching	Load Jump
Gholami et al. [14]		✓			✓	✓		✓	
Aligholian et al. [15]		✓			✓				✓
Vosughi et al. [17]		✓			✓			✓	✓
Ahmed et al. [19]		✓		✓	✓	✓		✓	
Gholami et al. [20]		✓	✓		✓	✓			
Ganjkhani et al. [21]		✓	✓		✓	✓		✓	
Ours	✓				✓	✓	✓	✓	✓

Note: I represents Voltage Magnitude from PMUs and Relays, II represents Complete Variables from PMUs, III represents Complete Variables from Relays, IV represents Network Data from Communication Network Logs.

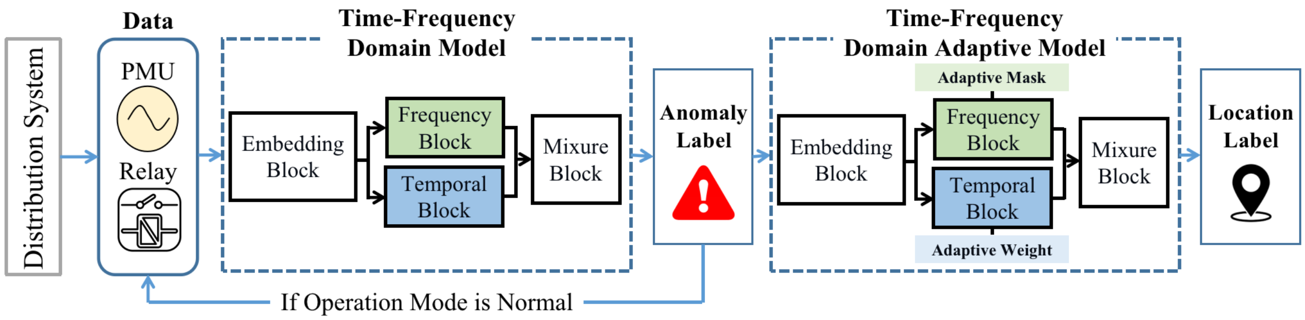

Fig. 1: The framework for anomaly detection and localization in distribution systems.

the performance instabilities observed in single-domain models such as [24], [25].

The remainder of this article is organized as follows. Section II introduces preliminaries. Section III presents the proposed TFDA. Section IV describes the setup of our case study. Section V provides the experimental results and analysis. Section VI concludes the paper with a summary of our work.

II. Preliminaries

This section provides a brief introduction to the proposed framework for anomaly detection and localization in distribution systems. It also describes the inputs, outputs, and types of anomalies considered in this work.

A. The Framework

The proposed framework is illustrated in Fig. 1. Measurement values are collected from various sensors in the distribution system and initially analyzed using the model without adaptive mechanisms, which is called time-frequency domain model (TFD). This model acts as an operational mode detector, which produces anomaly detection labels. If the detected operational mode is normal, it indicates the absence of anomalies, and localization module is not triggered. Data from subsequent time windows are continuously evaluated by the operational mode detector until an anomaly is identified. Once an anomaly is detected, the data would be handled by the corresponding TFDA to determine the anomaly's location labels.

B. Input and Output

This work only considers three-phase voltage magnitudes collected by two types of sensors: PMUs and relays, which sample with different time scale. Specifically, the PMU measurements sampled at a higher resolution R_h are denoted as $V_h = [V_{h1}, V_{h2}, V_{h3}, \dots, V_{hn}]$, where n represents the number of PMUs. The relay measurements sampled at a lower resolution R_l are denoted as $V_l = [V_{l1}, V_{l2}, \dots, V_{lm}]$, where m represents the number of relays. The first task of this work is to construct a mapping from the measurements to the anomaly label L_a, followed by constructing a mapping from the measurements to the anomaly localization label L_l:

$$L_a = f_{\text{TFD}}(V_h, V_l), \quad L_a \in \mathcal{A} \tag{1}$$

$$L_l = f_{\text{TFDA}(L_a)}(V_h, V_l), \quad L_l \in \mathcal{L}_{L_a}. \tag{2}$$

The models f_{TFD} and f_{TFDA} correspond to the two tasks, representing the time-frequency domain model and the time-frequency domain adaptive model, respectively. $\mathcal{A}$ represents the set of anomaly events, and $\mathcal{L}_{L_a}$ denotes the set of localization labels corresponding to the anomalies L_a. The specific experimental setup is described in the following sections.

C. Types of Anomaly

Building upon prior study, this work considers a more comprehensive set of anomaly events to better validate the adaptive performance of our model across different situations. Specifically, $\mathcal{A}$ includes six possible operational modes.

1) Normal Operation: When the distribution system operates under stable and normal conditions, it is assumed that no anomalies are present. Some prior works [17] included an additional 0/1 anomaly detection step before anomaly classification. In this study, we perform anomaly detection and classification simultaneously, thereby enhancing the efficiency of safety monitoring in the distribution system. This state is not associated with a localization label.

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2) Fault: A short-circuit fault occurs when a segment of the distribution system becomes shorted, which cause an instantaneous surge in current. This triggers the nearest protective relay to disconnect, which lead to a cascade of effects. In this work, we assume that a short-circuit fault can occur on any branch. Thus, the localization labels for short-circuit faults correspond to individual branch.

3) DER Switching: Sudden disconnection or connection of DER can cause significant fluctuations in the operational state of a distribution network. Timely identification of switching events allows relevant authorities to adjust scheduling decisions and prevent more severe consequences. In this study, we assume that switching events can occur at any DER connection point within the system. Therefore, the localization labels correspond directly to the connection points of the DERs.

4) Load Jump: Load jumps can also cause fluctuations in voltage and frequency during the operation of a distribution network. Significant load changes may place additional stress on switching equipment within the lines, potentially shortening their lifespan. In this study, the locations of variable loads are used as the labels for load jump anomalies.

5) Inverter Attack: For attacks on inverters, the typical approach is to tamper with the sensor measurements, which in turn affects the controller, leading to incorrect control commands and disrupting the operation of the inverter. Following the work [22], our attack model is set as follows:

$$M_A = \theta \cdot M + \epsilon \tag{3}$$

where M_A represents the attacked measurement values, while M represents the original measurement values. θ is the multiplicative factor matrix, and ϵ is the vector, both defining the unknown attack weights, with compatible dimensions. We assume that each inverter in the DERs may be subjected to an attack, and thus, the localization labels correspond one-to-one with the inverters at each location.

6) Relay Attack: In addition to attacks on inverters, this study also considers the possibility of false data injection attacks targeting protection relays. Tampering with relay measurements can cause noramlly operating inverters to detect abnormal voltage values, leading to unintended relay trips. Since the effects of such attacks are similar to those caused by short-circuit faults, it is crucial to distinguish between them. In this study, the localization labels for this type of attack correspond to the locations of the protection relays.

III. Time-Frequency Domain Adaptive Model

This work involves two models, TFD and TFDA, designed for anomaly detection and adaptive anomaly localization tasks, respectively. Since the TFD model is a simplified version of TFDA, with the adaptive mechanisms in both the time-domain and frequency-domain modules removed, we focus on introducing the TFDA to maintain clarity and conciseness in the article.

A. Overall Architecture

The framework is depicted in the Fig. 2. The input consists of data at different resolutions, which is preprocessed and transformed into tensor representations suitable for neural networks. The spatial patch embedding module is then applied to address the challenges posed by varying resolutions. The processed data is subsequently passed through the time-frequency domain module, which comprises two components: the adaptive frequency block (as shown on the left) and the adaptive temporal block (as shown on the right). Features from both blocks are combined and fed into the mixture convnet for integration. After passing through multiple layers of the time-frequency domain adaptive module, the data is processed by the projection layer, which maps it to the appropriate dimensionality for the task.

B. Spatial Patch Embedding

The primary purpose of the spatial patch embedding is to aggregate data from multi-resolution inputs while incorporating graph priors. The patch operation allows the model to focus not only on the information at individual time points but also to aggregate content over a period of time, thereby alleviating the information imbalance caused by multi-resolution data.

Since tensors from various resolutions cannot be directly concatenated, we first align the data with different resolutions along the real-world timeline. Then, the missing data on the low-resolution timeline is filled in using the most recent available data from the past relative to the target time point. Afterward, the extended data is concatenated with the high-resolution measurements to obtain $V \in \mathbb{R}^{3N_s \times L}$, Here, $N_s = n + m$ represents the total number of sensors, 3 represents the three-phase measurements from each sensor, and L denotes the unified time window length.

Then, we construct the adjacency matrix $\mathbf{A} \in \mathbb{R}^{N_s \times N_s}$ representing the relationships between different sensors. Based on the structure of the distribution system, we treat the location of each sensor as a graph node. If two sensors are connected by a line, and no other sensors lie in between them, we define that there is an edge between them. We incorporate the prior graph information into the temporal data through a graph convolutional network (GCN), as shown below:

$$Y_{\text{GCN}} = \mathbf{D}^{-\frac{1}{2}} \left(\mathbf{A} + \mathbf{I}_n \right) \mathbf{D}^{-\frac{1}{2}} V \theta \tag{4}$$

where $\mathbf{I}_n$ is a identity matrix, and $\mathbf{D}$ is the degree matrix. Subsequently, we segment each time series into patches based on [23]. The data Y_{GCN} is divided into N_p patches $[P_1, P_2, \ldots, P_{N_p}]$, where the size of each patch P is predefined. Meanwhile, the stride S is used to define the interval between each patch. Additionally, we incorporate standard operations for time series embedding, adding a learnable parameter $W_{pe} \in \mathbb{R}^{C_{\text{SPE}} \times N_p}$ for positional embedding, where C_{SPE} is the number of output channels of this module. The final output of the spatial patch embedding can be represented as follows:

$$Y_{\text{SPE}} = f_{emb}([P_1, P_2, \ldots, P_{N_p}]) + W_{pe}. \tag{5}$$

C. Time-Frequency Domain Adaptive Module

The data is processed by the time-frequency domain adaptive module, which serves as the core part of our model for

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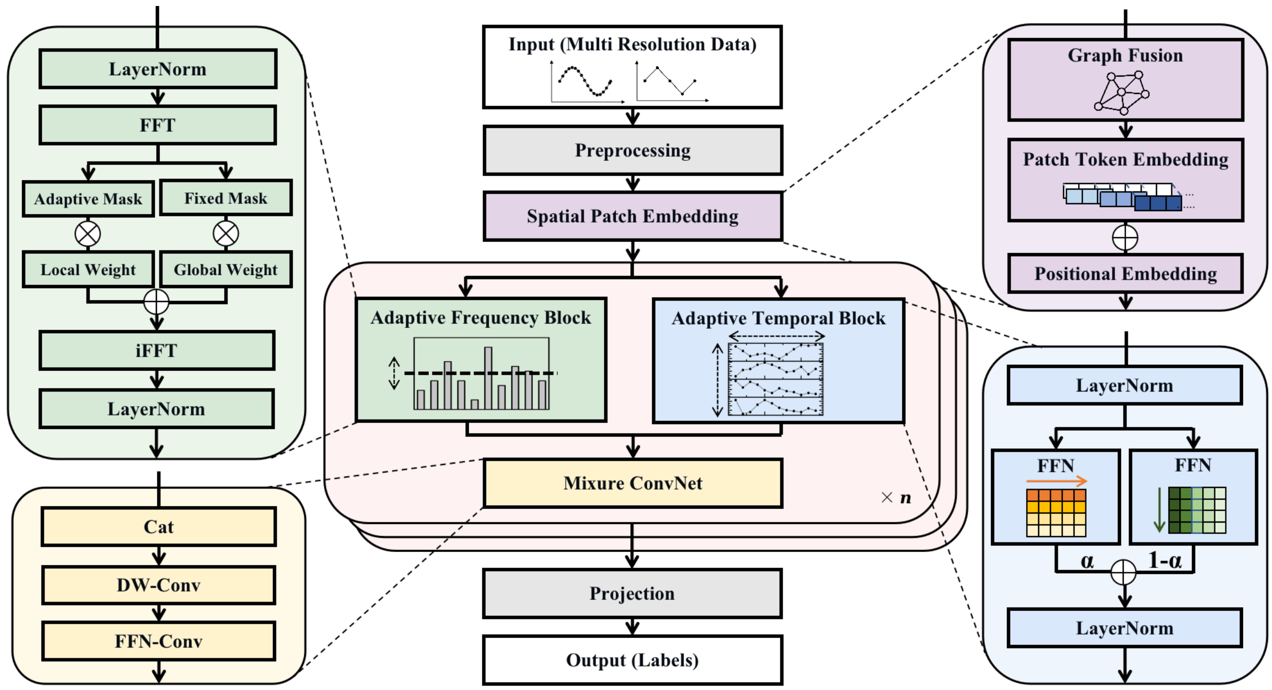

Fig. 2: The framework of time-frequency domain adaptive model.

anomaly detection and localization. This module consists of three components: the adaptive frequency block, the adaptive temporal block, and the mixture convnet. The time-frequency domain adaptive modules are connected sequentially, with the output of each layer serving as the input to the next. This data flow is formally defined as:

$$X_{\text{TFDB},i} = \begin{cases} Y_{\text{SPE}} & \text{if } i = 0 \\ Y_{\text{TFDB},i-1} & \text{if } i > 0 \end{cases} \tag{6}$$

where $X_{\text{TFDB,i}}$ represents the input to the i-th layer, and $Y_{\text{TFDB},i-1}$ represents the output of the $i-1$-th layer. For simplicity in subsequent explanations, the layer indices will be omitted.

1) Adaptive Frequency Block: First, we introduce the adaptive frequency block shown in green at the top left. We employ a common operation, where the Discrete Fourier Transform (DFT) is first used to map the features into the frequency domain for processing, followed by the Inverse Discrete Fourier Transform (IDFT) to map them back to the time domain for subsequent analysis [24]. In practice, these transforms are efficiently implemented using the Fast Fourier Transform (FFT) algorithm and its inverse (IFFT). These two symmetric operations can be expressed as:

$$F_k = \sum_{t=0}^{T-1} X_t \cdot e^{-j(2\pi/T)kt} \tag{7}$$

$$X_t = \frac{1}{T} \sum_{k=0}^{T-1} F_k \cdot e^{j(2\pi/T)kt} \tag{8}$$

where indices k and t correspond to the frequency domain and time domain, respectively. For $0 \leq k \leq T-1$, the DFT transforms the complex numbers $X_0, X_1, \ldots, X_{T-1}$ into a new sequence of complex numbers $F_0, F_1, \ldots, F_{T-1}$, while the IDFT performs the reverse operation.

In this block, we first obtain the frequency domain representation F of X_{TFDB} through the DFT operation. We then compute the power spectrum $P = |F|^2$, which serves as a reference for the strength of different frequency components. We only process the portion of the frequency spectrum with higher energy, as it contains more informative content, while the lower-energy components, such as high-frequency noise, are more likely to cause interference. We set a threshold γ as the boundary, and the corresponding mask can be expressed as follows:

$$Mask = F \odot (P > \gamma) \tag{9}$$

where $\odot$ represents element-wise multiplication. We define two forms of γ: one is a parameter that can be trained through gradient descent, and the other is a fixed parameter (which is directly set to 0 in our experiment). The former corresponds to a local mask M_{adaptive}, while the latter corresponds to a global mask M_{fixed}. Both are element-wise multiplied by their corresponding learnable weights and then summed together, resulting in comprehensive and adaptive information F_{sum}, which can be expressed as:

$$F_{\text{sum}} = M_{\text{adaptive}} \odot W_{\text{loacal}} + M_{\text{fixed}} \odot W_{\text{global}}. \tag{10}$$

Subsequently, the output of this block, Y_{AFB}, can be obtained through the IFFT operation.

2) Adaptive Temporal Block: Our module also focuses on information in the time domain. Generally, time-domain anomaly information can be categorized into two types: anomalies in the relationships between channels and anomalies in the trends of individual channels. Both types of anomaly information provide valuable insights for localization

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tasks. However, their impacts vary depending on the specific anomaly task, making it necessary to assign adaptive weights to achieve an optimal balance.

Based on the above idea, X_{TFDB} is processed through feedforward neural network (FFN) along two separate directions. After reshaping to unify the dimensions, a weighted summation is conducted using the adaptive weight α, yielding the final output Y_{ATB}:

$$f_{\text{FFN}}(\cdot) = f_{\text{linear}}(f_{\text{act}}(f_{\text{linear}}(\cdot))) \tag{11}$$

$$Y_{\text{ATB}} = \alpha f_{\text{FFN1}}(X_{\text{TFDB}}) + (1-\alpha) f_{\text{FFN2}}(X_{\text{TFDB}}^{T}) \tag{12}$$

where f_{FFN1} and f_{FFN2} correspond to independent FFN layers, each composed of simple linear layers f_{linear} and the nonlinear activation functions f_{act}. The superscript T denotes the transposition of the tensor along the channel and temporal dimensions. The adaptive weight α is initialized as a random value within the range $[0, 1]$ and is constrained during training to remain within this interval using the clamp operation.

3) Mixure ConvNet: For the feature information obtained from the temporal and frequency domain blocks, we employ depthwise convolution (DW-Conv) and pointwise convolution (PW-Conv) to comprehensively combine the feature information from both domains. In DW-Conv, each channel is processed with an individual convolutional kernel, ensuring that the number of channels remains unchanged before and after the operation. In PW-Conv, a 1×1 convolutional kernel is used, which maintain the feature dimensions throughout the process. This lightweight combination effectively captures and integrates features [25]. The steps of this block can be represented as follows:

$$Y_{\text{DWC}} = f_{\text{DWC}}([Y_{\text{AFB}}, Y_{\text{ATB}}]) \tag{13}$$

$$Y_{\text{TFDB}} = f_{\text{FFNC}}(Y_{\text{DWC}}) = f_{\text{PWC1}}(f_{\text{act}}(f_{\text{PWC2}}(Y_{\text{DWC}}))) \tag{14}$$

where f_{DWC} and f_{PWC} represent DW-Conv and PW-Conv, respectively. Additionally, f_{FFNC} organizes the PW-Conv in a structure similar to a FFN. The output Y_{TFDB} from the final layer of the time-frequency domain adaptive module is then passed to the projection layer, where it is mapped to the label dimension associated with the specific task.

IV. Experimental Preparation

A. Datasets

We conduct experiments using a real-time Typhoon hardware-in-the-loop (HIL) testbed to validate the effectiveness of the proposed method. As shown in Fig. 3, two HIL 602+ emulators are connected to the simulation environment on a computer via USB/Ethernet. The automated simulation runs are implemented through TyphoonTest, which generates operational data for different anomaly scenarios. Subsequently, a detailed description of the simulation setup is provided below.

As shown in Fig. 4, the testing environment is based on the original IEEE 33-node system with some modifications, based on the work in [21]. Wind turbines were added to buses 14, 18, 22, 33, and energy storage systems were added to buses 18 and 33. The wind speed data used in this setup were sourced from

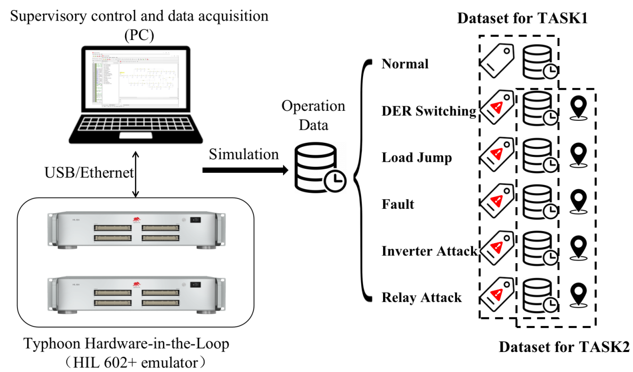

Fig. 3: Overview of the HIL-based simulation process and dataset construction.

TABLE II: Overview of the Dataset.

Anomaly	Location Label	Number
DER Switching	Bus 4, 18, 22, 33	16000
Load Jump	Bus 3, 16, 24, 28	16000
Fault	All the branches	48000
Inverter Attack	Bus 4, 18, 22, 33	16000
Relay Attack	Branches (1-2), (12-13), and (28-29).	14000

Bureau of Meteorology Australia [26]. As mentioned in previous work [27], the integration of renewable energy resources can affect the accuracy of anomaly detection. Therefore, we employed this wind speed data to create a more challenging scenario. Protection relays were installed on branches (1-2), (12-13), and (28-29), while PMUs were installed at buses 4, 6, 11, 17, 21, 24, and 32. The data acquisition frequency for the protection relays, R_l, is 10 datapoints/second, while for the PMUs, R_h is 100 datapoints/second. The detailed original system parameters are provided in [28].

For the various anomalies, we assume that short-circuit faults can occur on any branch in the system, DER switching events may occur at buses 14, 18, 22, and 33, and load jumps are assumed to occur at buses 3, 16, 24, and 28. Additionally, we assume that the attacker only target one location at a time, with inverter attacks potentially occurring at buses 14, 18, 22, and 33 and relay attacks on branches (1-2), (12-13), and (28-29). A detailed breakdown of the anomaly types, their specific locations, and the number of generated samples for each category is provided in Table II.

Based on the simulation scenario described above, we conduct experiments with a simulation step size of 400 μs. Each simulation begins with ensuring the distribution system operates under normal conditions, after which a single anomaly is introduced by altering simulation variables. It is worth noting that simultaneous anomaly location classes are not included in our target classes, as our framework utilizes high-resolution data and operates on a small time window to locate and classify anomalies. Given such a small timescale of our detection window, the probability of two independent anomalies initiating at the exact same instant is considered negligible. We collect 1.2 seconds of data for each simulation, with the anomaly occurring at the midpoint. To better reflect real-world scenarios where the exact location of anomalies is typically unknown, we segment the time series into 900-

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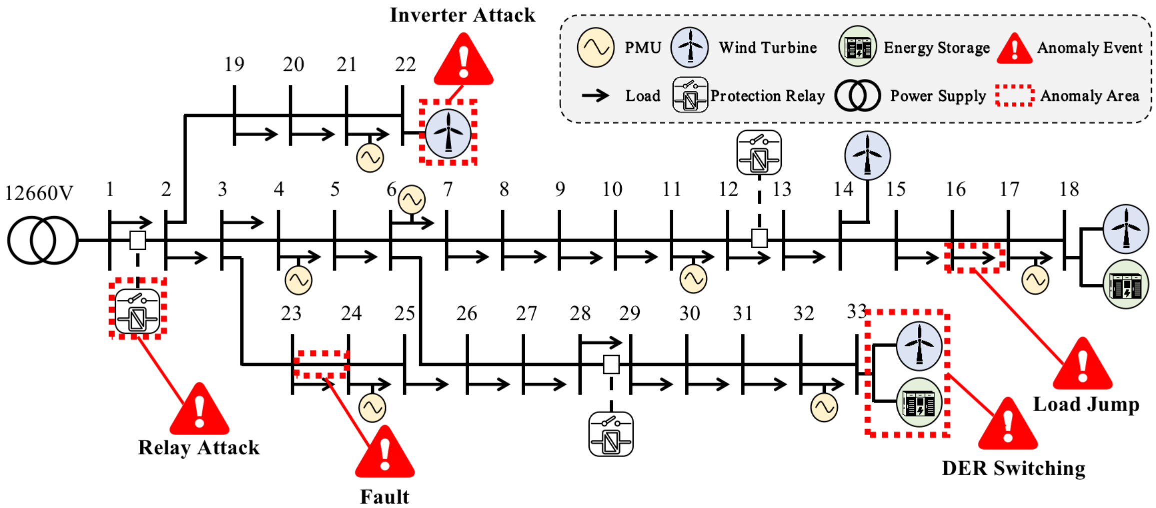

Fig. 4: Modified IEEE 33-bus test feeder.

millisecond windows, ensuring that the anomaly occurs within the window but not too close to its edges. As shown in the figure, we construct the dataset for task 1 by pairing the time-windowed data with the anomaly labels, and for task 2 by pairing time-windowed data, excluding normal operations, with the anomaly location labels.

B. Experimental Setting

1) Baseline: In this experiment, we first compare baselines using recently proposed methods for anomaly detection and localization in power distribution systems, including LSTM [21] ,CNN [16], SDE [9] and GAT [10]. Additionally, we evaluate time-series analysis methods that have gained significant attention in recent top-tier computer science conferences, including DLinear [29], Autoformer [30], FiLM [24], PatchTST [23], Crossformer [31], TimesNet [32], and ModernTCN [25]. Due to differences in experimental setups and the challenge to fully reproduce previous data integration methods with comprehensive variable data, we adopt the data integration approach proposed in this work for comparison.

2) Metrics: We evaluate the performance of different methods using four metrics: Recall, Precision, F1-score, and Accuracy. Here, TP, TN, FP, and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively. These metrics are defined as follows:

$$\begin{cases} \text{Recall} &= \frac{\text{TP}}{\text{TP+FN}} \\ \text{Precision} &= \frac{\text{TP}}{\text{TP+FP}} \\ \text{F1-score} &= \frac{2\times\text{TP}}{2\times\text{TN+FP+FN}} \\ \text{Accuracy} &= \frac{\text{TP+TN}}{\text{TP+TN+FP+FN}} \end{cases} \tag{15}$$

3) Loss Function: In this work, we use the cross-entropy loss function, which is commonly employed loss function for classification problems. The model's optimal parameters are obtained by minimizing this loss through gradient descent. The loss function can be expressed as follows:

$$Loss(y, \hat{y}) = -\sum_{p=1}^{P}\sum_{q=1}^{Q} y_{p,q} \log \hat{y}_{p,q} \tag{16}$$

TABLE III: Performance Comparison of Recall(%), Precision(%), F1-Score(%), and Accuracy(%) for Anomaly Detection

Method	Precision	Recall	Accurracy	F1-score
CNN [21]	91.09	93.92	93.93	92.48
LSTM [16]	90.70	95.21	94.17	92.90
Dlinear [29]	89.20	92.36	92.57	90.75
Autoformer [30]	82.93	82.53	83.83	82.73
FiLM [24]	89.46	89.86	91.47	89.66
PatchTST [23]	88.68	90.04	91.33	89.35
SDE [9]	88.84	90.86	91.83	89.29
Crossformer [31]	93.03	93.59	94.50	93.31
TimesNet [32]	<u>95.18</u>	<u>96.33</u>	<u>96.40</u>	<u>95.75</u>
GAT [10]	94.85	95.39	95.70	95.11
ModernTCN [25]	94.94	94.93	94.17	94.93
TFD (Ours)	**99.09**	**99.20**	**99.30**	**99.14**

where P represents the total number of samples, Q denotes the number of classes, and y is the true class distribution, represented in one-hot encoded format. $\hat{y}_{p,q}$ represents the predicted probability of the p-th sample belonging to the q-th class, obtained using the softmax function, satisfying the condition $\sum_{q=1}^{Q} \hat{y}_{p,q} = 1$.

4) Training Settings: Our experiments are implemented in Python 3.8.19 using pytorch 2.4.0 libraries on a PC equipped with an Intel i7-8750H CPU, NVIDIA GTX 1060Ti GPU, and 16.0-GB RAM. We split the data set into three sections for training, validation, and testing with a ratio of 70:15:15. We use the RAdam optimizer with an initial learning rate of 0.001 for baselines and our method. We set the batch size to 32 and maximum training epochs to 100, with a patience of 10 for early stopping strategy. For our method, the dropout rate in the model is set to 0.1. In the spatial patch embedding, the size of each patch P is set to 10, and the stride S is set to 5. We configure the number of time-frequency domain adaptive modules to be 4, with f_{act} set to the GeLU activation function. In the mixture convnet, the hidden layer dimensions are set to three times the input dimensions.

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TABLE IV: Performance Comparison of Recall(%), Precision(%), F1-Score(%), and Accuracy(%) for Anomaly Localization

Metrics / Methods	Average Performance				The Best Performance				The Worst Performance			
	Precision	Recall	Accurracy	F1-score	Precision	Recall	Accurracy	F1-score	Precision	Recall	Accurracy	F1-score
CNN [21]	90.25	90.59	90.34	90.42	95.47	95.46	95.44	95.46	75.04	76.14	75.03	75.59
LSTM [16]	92.37	92.87	92.35	92.62	95.69	95.73	95.66	95.71	84.89	86.25	84.88	85.56
Dlinear [29]	89.28	89.92	89.28	89.59	95.52	95.60	95.50	95.56	83.42	85.12	83.53	83.26
Autoformer [30]	89.87	89.88	89.80	89.87	95.93	95.87	95.92	95.87	83.79	84.17	83.98	83.98
FiLM [24]	90.11	90.18	90.07	90.15	92.32	92.20	92.17	92.26	86.72	86.64	86.77	86.68
PatchTST [23]	90.96	91.14	90.88	91.05	96.34	96.28	96.28	96.31	80.54	81.46	80.50	81.00
Crossformer [31]	94.01	94.28	94.31	94.15	97.11	97.05	97.06	97.08	88.23	89.34	88.25	88.78
SDE [9]	92.15	92.87	92.23	92.42	97.11	97.14	97.10	97.13	86.75	88.87	86.72	87.80
TimesNet [32]	92.09	92.45	92.16	92.27	97.21	97.15	97.17	97.18	78.79	80.46	78.69	79.62
GAT [10]	95.66	94.84	94.70	94.70	96.73	96.81	97.02	96.77	91.95	92.35	91.91	92.04
ModernTCN [25]	95.19	95.17	95.19	95.18	96.77	96.70	96.72	96.73	93.19	93.22	93.38	93.20
TFDA (Ours)	**97.94**	**97.90**	**97.95**	**97.92**	**99.16**	**99.17**	**99.16**	**99.16**	**97.19**	**97.08**	**97.09**	**97.10**

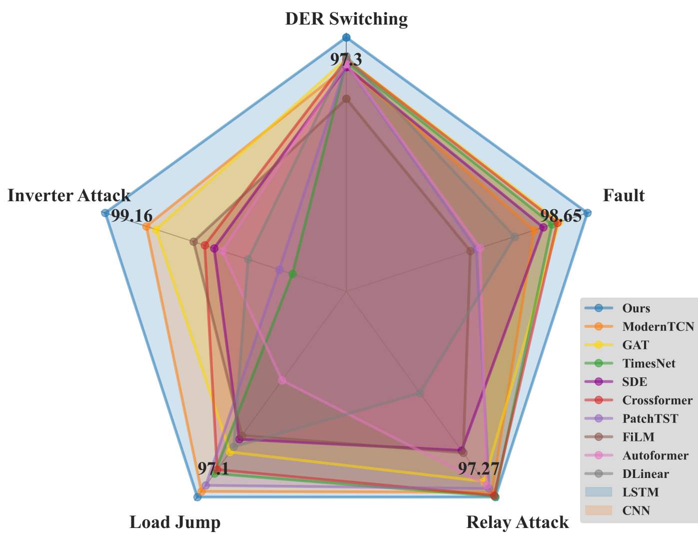

Fig. 5: Comparison between models on different anomaly localization tasks.

V. Experimental Result

A. Baseline Comparasion

We validate the effectiveness of the proposed TFD on the first task, focusing on anomaly detection. The overall experimental results are shown in Table III, where the bolded data represents the optimal results among the different methods, while the underlined data represents the second-best results. It can be observed that our method outperforms the second-best method, achieving improvements of 3.91%, 2.87%, 2.90%, and 3.39% in Precision, Recall, Accuracy, and F1-score, respectively.

We validate the effectiveness of the proposed TFDA module on the second task, focusing on anomaly localization. The overall experimental results are summarized in Table IV. Unlike the previous task, this task comprises multiple subtasks corresponding to different anomaly types. To provide a comprehensive evaluation, we report the average, best, and worst performance across these subtasks. Our method demonstrates a consistent average improvement over the second-best baseline, achieving gains of 2.75% in Precision, 2.73% in Recall, 2.76% in Accuracy, and 2.74% in F1-score.

The detailed per-subtask results in Fig. 5 provide further insights into the robustness of our approach. Notably, the performance profile of our method consistently envelopes those of all other baselines, demonstrating its broad superiority across diverse anomaly types. Moreover, our approach exhibits the smallest performance variance, with the narrowest gap between best and worst cases, highlighting its exceptional stability. Remarkably, even the worst-case performance of our method remains competitive with or surpasses the best-case performance of other methods, confirming the practical reliability and high lower-bound performance of our proposed TFDA module.

B. Ablation Experiments

We conduct ablation studies to validate the necessity of each module within the overall model. By removing specific modules from the complete model, we observe the resulting performance degradation. We evaluate the performance of the main modules, including spatial patch embedding, temporal block, and frequency block, followed by the assessment of the two adaptive operations, temporal adaptation and frequency adaptation. The key experimental results are shown in the fig. 6. The abbreviations "w/o SPE," "w/o TB," and "w/o FB" represent the removal of the corresponding main modules, while "w/o TA" and "w/o FA" indicate the removal of the corresponding adaptive operations.

It can be observed that the main modules have significant impacts on the model, with the data aggregation module spatial patch embedding having the largest effect, which shows a greater than 2% impact on all metrics. The experiments with "w/o TA" and "w/o FB" also validate that the adaptive operations play a crucial role. Furthermore, the longer error bars in the "w/o TA" and "w/o FB" experiments suggest that these adaptive operations contribute to the model's stability across various tasks.

At the same time, we also validate that the inclusion of these modules does not lead to a significant increase in the number of parameter or inference time. The experimental results are shown in Fig. 7. The number of parameters corresponds to the average value for each anomaly in the ablation study, while the inference time is measured with a batch size of 1000. For clearer comparison, the radius of each circle representing the model is proportional to the square of the number of

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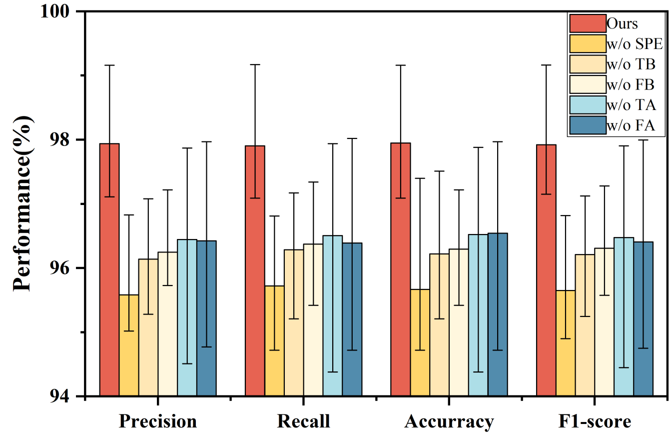

Fig. 6: Visual comparison of performance metrics in ablation experiments.

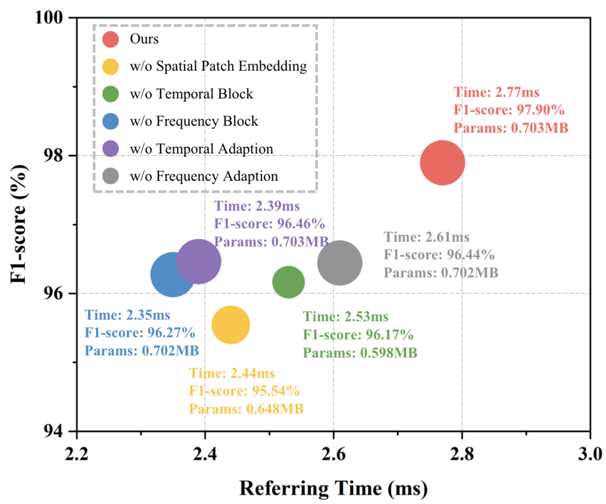

Fig. 7: Comparison regarding the model performance, inferring time, and the number of model parameters.

parameters. It can be observed that with the removal of each module, both the number of parameter and inference time change within an acceptable range, without affecting the model's deployability and practicality. This further validates the effectiveness of the individual modules.

C. Analysis and Visualization

We explore the impact of hyperparameters on model performance through experiments. The hyperparameters considered include the number of time-frequency domain adaptive modules, patch size and stride in spatial patch embedding, and the hidden layer dimension in the mixture convnet. The model performance represents the average results of TFDA across different anomaly localization tasks. The experimental results are shown in the Table V, where the "hidden layer dimension ratio" represents the ratio of the hidden layer dimension to the input dimension. The table shows the standard deviation of results from multiple repeated experiments under specific hyperparameters, rather than the standard deviation of different task results from a single experiment. We observe that the model exhibits robustness under different hyperparameter settings. This further proves that the superiority of our method is not due to an increase in parameters, but rather because of the clever architecture and feature design. In practical applications, we can adopt the optimal hyperparameters based on the experimental results to prevent model overfitting.

To explore the model's adaptability to the anomaly localization task, we visualize the adaptive components using heatmaps, as shown in Fig. 8. In the figure, part (a) depicts the adaptive parameter γ in the frequency block, which represents the energy spectrum threshold considered in the frequency domain, while part (b) shows the adaptive parameter α in the temporal block, which represents the weight ratio between inter-channel information and single-channel information in the time domain.

First, we conduct a horizontal analysis of γ and α across different layers within the same task. Overall, γ exhibits larger values in the last two layers compared to the first two, suggesting that the later layers impose stricter constraints on the information, enabling the extraction of more detailed features than the earlier layers. For α, we observe significant differences across layers, with values ranging from near 0 to near 1. This suggests that both types of anomaly information in the time domain are explored to varying extents in different layers.

Next, we perform a vertical analysis of the adaptive parameters across different tasks. We observe that the variations in the adaptive parameters exhibit distinct characteristics for each task. For instance, in the localization of inverter attacks, the values of γ are relatively uniform, whereas in the DER switching localization task, γ shows greater variation. In the fault localization task, α tends to have higher values in the last two layers, while in the inverter attack localization task, higher α values are observed in earlier layers. These findings indicate that the adaptive parameters of the model vary across different localization tasks, reflecting different emphases on anomaly information. The adaptive operations enable the model to maintain stable performance when dealing with various types of anomalies, thereby better meeting the safety requirements of the power distribution system.

D. Supplementary Validation

As shown in the table VI, we conducted additional validation experiments by incorporating three-phase current measurements into our method, thereby enriching the information input. The results indicate that while the inclusion of current data does lead to a modest improvement in performance, the enhancement is not substantial. This outcome aligns with our expectations, since the proposed method is specifically designed for scenarios with limited variable availability and thus does not deeply model inter-variable correlations. Consequently, the addition of extra variables does not yield significant gains. More importantly, these results demonstrate that our approach maintains strong performance even under variable-constrained conditions, ensuring high reliability in safety-critical applications.

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TABLE V: Impact of Hyperparameters on Model Performance

Hyperparameters	Values	Precision	Recall	Accurracy	F1-score
Number of main modules	1	97.28 ± 0.03	97.28 ± 0.03	97.28 ± 0.04	97.28 ± 0.03
	2	97.16 ± 0.02	97.15 ± 0.03	97.18 ± 0.02	97.15 ± 0.03
	3	97.60 ± 0.05	97.59 ± 0.05	97.59 ± 0.05	97.60 ± 0.05
	4	**97.94 ± 0.02**	**97.90 ± 0.03**	**97.95 ± 0.02**	**97.92 ± 0.02**
	5	97.53 ± 0.02	97.54 ± 0.03	97.55 ± 0.02	97.53 ± 0.03
Patch size and stride	5/3	95.94 ± 0.06	95.98 ± 0.07	95.97 ± 0.04	95.95 ± 0.06
	5/5	97.18 ± 0.02	97.25 ± 0.03	97.20 ± 0.05	97.22 ± 0.02
	10/5	**97.94 ± 0.02**	**97.90 ± 0.03**	**97.95 ± 0.02**	**97.92 ± 0.02**
	10/10	96.77 ± 0.05	96.82 ± 0.07	96.84 ± 0.06	96.78 ± 0.06
Hidden layer dimension ratio	1	97.41 ± 0.03	97.43 ± 0.02	97.44 ± 0.05	97.41 ± 0.02
	2	**97.94 ± 0.02**	**97.90 ± 0.03**	**97.95 ± 0.02**	**97.92 ± 0.02**
	3	97.11 ± 0.02	97.14 ± 0.04	97.14 ± 0.03	97.10 ± 0.03

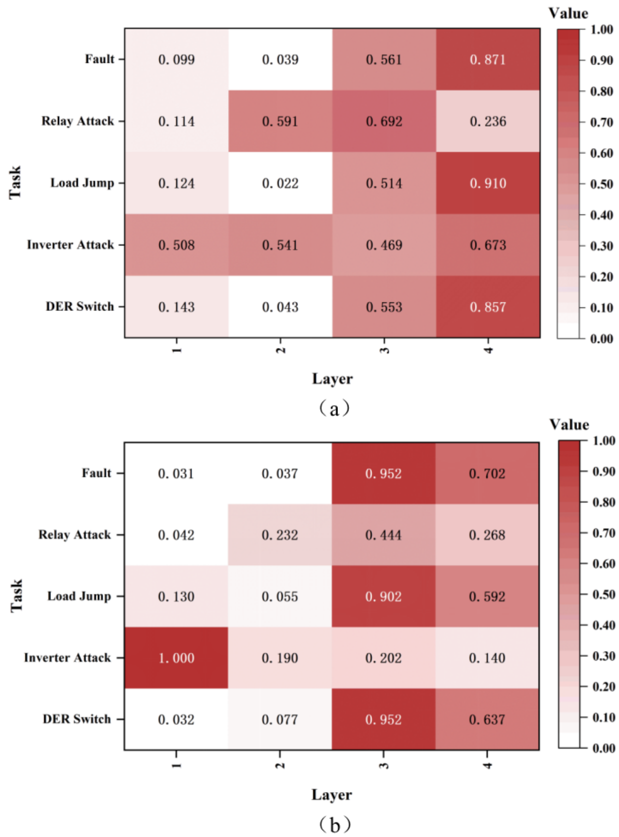

Fig. 8: Visualization of adaptive parameters in the model.

TABLE VI: Impact of Additional Measurements on Performance.

Method	With Current			Without Current		
	Average	Best	Worst	Average	Best	Worst
Precision	97.97	99.19	97.19	97.94	99.16	97.19
Recall	97.95	99.20	97.15	97.90	99.17	97.08
Accuracy	97.99	99.20	97.13	97.95	99.16	97.09
F1-score	97.96	99.19	97.17	97.92	99.16	97.10

VI. Limitation

1) Practical Data Constraints: A primary limitation of this study is the reliance on simulated datasets, necessitated by the strict data confidentiality and privacy constraints associated with actual distribution networks. While the simulation environment was carefully designed to approximate practical operating conditions, it may not fully capture the intricate stochasticity and measurement non-idealities inherent in real-world environments. Future efforts will focus on collaborating with utility operators to leverage real-world field data for further validation and fine-tuning, thereby enhancing the model's reliability and facilitating its practical deployment.
2) Concurrent Anomaly Resolution: Given the statistical improbability of multiple independent anomalies occurring simultaneously within our timeframe, this study focuses on identifying a single primary event per detection window. Consequently, the current framework is not explicitly designed to disentangle overlapping signatures of concurrent events. However, recognizing the theoretical importance of such scenarios, future research will focus on extending the models capability through multi-label learning architectures to handle rare but complex multi-anomaly interactions.
3) Generalization to Low-Current and Unknown Anomalies: The proposed framework faces inherent challenges in achieving high-precision localization for low-current anomalies, such as high-impedance faults (HIFs). In our variable-limited scenario, the current deviations induced by HIFs are often too subtle to provide a distinct spatial signature across the network. Furthermore, as the model operates under a closed-set assumption, it is fundamentally unable to classify or localize unknown anomaly types that were not present in the training distribution. Future research will explore the transition toward an open-set recognition (OSR) paradigm. By incorporating uncertainty estimation or self-supervised contrastive learning, future iterations aim to enhance the detection sensitivity to near-threshold signals and provide a more robust situational awareness system.

VII. Conclusion

In this paper, we address the problem of adaptive anomaly detection and localization in distribution systems using only voltage magnitude. Specifically, we propose a data-driven time-frequency domain adaptive model. The model incorporates a spatial patch embedding mechanism, which processes time series data as patches, mitigating the impact of multi-resolution information while integrating graph priors based on the relative positions of sensors. For frequency-domain information, an adaptive threshold filters out low-energy components in the spectrum, and for temporal information, adaptive weights balance inter-channel and intra-channel contributions. Extensive experiments demonstrate the effectiveness of our approach, and ablation studies validated the necessity of the proposed innovative blocks. In the future, we aim to explore the model's generalization to unseen anomalies and anomaly locations.

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